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Patent Public Search | Text View

United States Patent Application Publication

20250284764

Kind Code

A1

Publication Date

September 11, 2025

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FFT ENGINE HAVING COMBINED BIT-REVERSAL AND MEMORY TRANSPOSE OPERATIONS

Abstract

An example system includes logic circuitry to receive digital signals and perform transform operations to generate N output samples for each of the digital signals, k output samples at a time, where k is an integer of 2 or greater and N is an integer multiple of k. Such system further includes circular shift circuitry that includes data multiplexers and address multiplexers to write respective sets of k output samples of the N output samples to k banks of a first memory. For each set of k output samples, the address multiplexers are used to generate k addresses that are bit-reversed with respect to the k output samples, and write a respective one of the k output samples at an indexed ordered position in a respective one of the k banks. The output samples are then read in linear address order from the k banks of the first memory and stored in transposed format in a second memory.

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Family ID: 71072653

Appl. No.: 19/214240

Filed: May 21, 2025

Related U.S. Application Data

parent US continuation 17331215 20210526 parent-grant-document US 11734382 child US 18447029

parent US continuation 16221470 20181215 parent-grant-document US 11170071 child US 17331215

parent US division 18447029 20230809 parent-grant-document US 12346402 child US 19214240

Publication Classification

Int. Cl.: G06F17/14 (20060101); G06F7/76 (20060101); G06F13/28 (20060101)

U.S. Cl.:

CPC G06F17/142 (20130101); G06F7/768 (20130101); G06F13/287 (20130101);

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] The present U.S. Patent Application is a divisional of U.S. patent application Ser. No. 18/447,029, filed Aug. 9, 2023, which is a continuation of U.S. patent application Ser. No. 17/331,215, filed May 26, 2021 (now U.S. Pat. No. 11,734,382), which is a continuation of U.S. patent application Ser. No. 16/221,470, filed Dec. 15, 2018 (now U.S. Pat. No. 11,170,071), each of which is incorporated by reference herein in its entirety.

BACKGROUND

[0002] Spectral analysis of electrical sense signals is useful in many scenarios. An example scenario involves spectral analysis of sense signals or of different signals (e.g., a received sense signal versus a target signal) to sense variations in a physical environment. Another example scenario involves spectral analysis of image signals. Radar systems are one example of system that relies on spectral analysis of sense signals. One example technique to perform spectral analysis analyze characteristics of the object(s) of sense signals involves performing Fast Fourier Transform (FFT) operations on the received signal, where the FFT operations generate output samples as a function of frequency.

[0003] Multi-dimensional Fast Fourier transforms are generally computed by taking one-dimensional (1D) Fast Fourier transforms along all dimensions, successively. As an example, by analyzing the FFT output samples over multiple dimensions, a Frequency-Modulated Continuous Wave (FMCW) radar system can detect the characteristics of an object related to the radar system. For example, three-dimensional (3D) FFT processing is used to determine an object's distance from the radar system, an object's velocity relative to the radar system, and an object's angle relative to the radar system.

SUMMARY

[0004] In accordance with at least one example of the disclosure, a data processing device comprises logic circuitry configurable to receive a plurality of digital signals and perform transform operations to generate N output samples for each of the plurality of digital signals, k output samples at a time, in which k is an integer of 2 or greater and N is an integer multiple of k. The data processing device further comprises circular shift circuitry that includes data multiplexers and address multiplexers configurable to write respective sets of k output samples of the N output samples to k banks of a first memory device. For each set of k output samples, the circular shift circuitry is further configurable to generate, using the address multiplexers, k addresses that are bit-reversed with respect to the k output samples, and write a respective one of the k output samples at an indexed ordered position in a respective one of the k banks. The data processing device is configurable to read the output samples in linear address order from the k banks of the first memory device and store the output samples in a transposed format in a second memory device.

[0005] In accordance with at least one example of the disclosure, the data processing unit includes an FFT engine that is embodied on an integrated circuit that also includes radar transmission and receiving circuitry. The receiving circuitry may include one or more analog-to-digital converters

(ADCs) configurable to digitize received radar signals for input to the FFT engine. The integrated circuit may further include processing circuitry coupled to receive the output of the FFT engine. [0006] In accordance with at least one example of the disclosure, a method for processing digital signals is provided, in which the method operations may be commensurate with the processing described herein. [0007] Other configurations, aspects and features are described below.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] For a detailed description of various examples, reference will now be made to the accompanying drawings in which:

[0009] FIG. 1 is a block diagram showing a prior art Fast Fourier Transform (FFT) engine for a system implementing multi-dimensional FFT;

[0010] FIG. 2 is a block diagram showing another prior art FFT engine for a system implementing multi-dimensional FFT;

[0011] FIG. 3 is a block diagram showing a radar system in accordance with various examples;

[0012] FIG. 4 is a block diagram showing part of a radar data processing system in accordance with various examples;

[0013] FIG. 5 is a block diagram showing an FFT engine in accordance with various examples;

[0014] FIGS. 6A-6D are block diagrams showing example operations of the FFT engine of FIG. 5 in accordance with various examples;

[0015] FIG. 7 is a block diagram of a radar data processor device in accordance with various examples; and

[0016] FIG. 8 is a flowchart showing an FFT engine method in accordance with various examples.

DETAILED DESCRIPTION

[0017] Disclosed herein are data processor devices, systems, and methods, involving a Fast Fourier Transform (FFT) engine topology with combined bit-reversal and memory transpose operations. As used herein “combined” bit-reversal and memory transpose operations” refers to performing FFT output sample transpose operations and performing FFT output sample bit-reversed ordering operations to store FFT output samples in data memory without incurring extra latency for a separate transpose operation. By using an FFT engine topology that performs at least some combined bit-reversal and memory transpose operations, data processing latency and the FFT engine footprint (integrated circuit area) is reduced compared to other FFT engine topologies, where FFT output samples are stored or used in a transposed order. Without limitation to other examples, the disclosed FFT engine topologies are relevant to multi-dimensional FFT analysis, where the efficiency of transpose operations affects each dimension.

[0018] FIG. 1 shows a block diagram of an example FFT engine **100** that can be used for multi-dimensional FFT analysis (e.g., for radar data processing or other sense signal analysis). For the FFT engine **100**, a 4X-FFT **101** is used. As used herein, a “KX-FFT” refers to a parallel pipelined FFT architecture to achieve K times parallelism by processing K inputs and outputting K outputs at a time. If K=4, the FFT is a 4X-FFT such as the 4X-FFT **101**. In some real-time applications, where high throughput is desired, several samples of an input sequence are processed in parallel and multiple outputs are provided in parallel. In the example 4X-FFT **101**, the FFT output samples $X[k]$, $X[k+N/4]$, $X[k+N/2]$, $X[k+3N/4]$ are generated from FFT input samples $x[4m]$, $x[4m+1]$, $x[4m+2]$, $x[4m+3]$, where $k=0, 1, \dots, N-1$, where $m=0, 1, \dots, N-1$, and where N is the size of the FFT. In the example of FIG. 1, an FFT of size $N=16$ is considered, where FFT input samples are input four at a time (samples 0, 1, 2, 3, then samples 4, 5, 6, 7, and so on) to the 4X-FFT **101** as represented in table **102**, and where the FFT output samples are output four at a time from the 4X-

FFT **101** and are bit-reversed as represented in table **103** (samples 0, 8, 4, 12, then samples 2, 10, 6, 12, and so on).

[0019] The FFT output samples are stored in local memory **110** with a multi-row-multi-column format as represented in table **111** using bit-reversed addressing **105**, which uses knowledge of the bit-reversal behavior of the 4X-FFT **101** to write the output samples from the 4X-FFT **101** in the multi-row-multi-column format of table **111**. The contents of the local memory **110** are stored in a data memory **120** with the format represented in table **121** using a direct memory access (DMA) logic **131**, where the same multi-row-multi-column format is used for each set of FFT output samples. The contents of the data memory are again read out in transposed order to perform a second dimensional FFT. The transpose read operation for 4 data samples consumes 4 clock cycles since they reside in different memory locations, thus reducing the effective throughput and speed of the 4X FFT engine **100**. The local memory **110** may be smaller in size than the data memory **120**. In some scenarios, the local memory **110** will be used to store one 1D-FFT outputs whereas the data memory **120** is used to store the outputs of multiple 1D-FFT outputs across multiple dimensions. The operations described for the FFT engine **100** of FIG. **1** are part of a larger data processing chain, where large data processing latency introduced for transpose operations of the FFT engine **100** are undesirable.

[0020] FIG. **2** shows a block diagram of another FFT engine **200** that can be used for performing multi-dimensional FFTs (e.g., for radar data processing or other sense signal analysis). For the FFT engine **200**, a 4X-FFT **201** is used, where FFT output samples $X[k]$, $X[k+N/4]$, $X[k+2N/4]$, $X[k+3N/4]$ are generated from FFT input samples $x[4m]$, $x[4m+1]$, $x[4m+2]$, $x[4m+3]$. In the example of FIG. **1**, an FFT of size $N=16$ is considered, where FFT input samples that are input four at a time (samples 0, 1, 2, 3, then samples 4, 5, 6, 7, and so on) to the 4X-FFT **201** as represented in table **202**, and where the FFT output samples are output four at a time from the 4X-FFT **201** in bit-reversed address order as represented in table **203** (samples 0, 8, 4, 12, then samples 2, 10, 6, 12, and so on). The FFT output samples for multiple FFTs are stored in local memory **210** with a multi-row-multi-column format as represented in table **211** using bit-reversal addressing **205**, which uses knowledge of the bit-reversal behavior of the 4X-FFT **201** to provide the multi-row-multi-column format. The FFT engine **200** also includes a transpose engine **220** and a transpose memory **221**, where the transpose memory **221** stores samples obtained from the local memory **210** in a transposed format represented in table **222**. The contents of the transpose memory **221** are stored in a larger data memory **230** with the format represented in table **222** using DMA logic **241**. The contents of the data memory **230** can be directly used to perform a second dimensional FFT since the data is already stored in a transpose order using transpose engine **220**. The local memory **210** may be smaller in size compared to data memory **230**. The operations described for the FFT engine **200** of FIG. **2** are part of a larger data processing chain, where data processing latency introduced by transpose operations of the FFT engine **200** is undesirable.

[0021] In some examples, an FFT engine that overcomes the latency issues discussed for the FFT engines **100** and **200** includes FFT logic configured to receive a plurality of digitized sense signals sequentially. As an example, each digitized sense signal includes radar chirp and radar chirp reflection information. In other examples, each digitized sense signal includes image information, or other sense signal information. In some examples, the FFT logic generates K FFT output samples at a time and N total FFT output samples for each digitized sense signal, where N is a multiple of K , and where K is an integer equal to at least 2. In one example N is 16, and K is 4. The FFT engine also includes: 1) a first memory device with a plurality of banks (e.g., K banks); and 2) a bit-reversed address generator and first set of circular shift components configured to shift between the plurality of banks when writing the generated FFT output samples in bit-reversed address order to the first memory device. The FFT engine also includes a second set of circular shift components configured to shift between the plurality of banks when reading FFT output samples in linear address order from the first memory device for storage in the second memory

device, where the first and second set of circular shift components together are used to read FFT output data in transpose order, and where at least some bit-reversal operations and at least some memory transpose operations are combined

[0022] In some examples, the disclosed FFT engine is part of a radar data processing device (e.g., an integrated circuit, chip, or multi-die module). An example radar data processing device includes or couples to at least one antenna configured to transmit chirp signals and to receive chirp reflection signals. Each transmitted chirp signal and related chirp reflection signals are combined by the radar data processing device and digitized as an input signal to the FFT engine, to produce FFT output samples stored in the second memory device. A DMA can also be included in the system that can be configured to read data in another dimension from the second memory device to the FFT engine to implement multi-dimensional FFTs where number of dimension is at least equal to 2

[0023] The radar data processing device also includes a processor configured to use the multi-dimensional FFT output samples stored in the second memory device to determine object parameters such as at least one of an object range, an object velocity, and an object angle relative to a chirp origin. With the disclosed FFT engine, data processing latency of devices implementing multi-dimensional FFT is improved relative to the data processing devices that use the FFT engines introduced in FIGS. 1 and 2. To provide a better understanding, various FFT engine options, data processing device options and/or other options are described using FIGS. 3-7 as follows.

[0024] FIG. 3 is a block diagram of a radar system 300 in accordance with various examples. In the example of FIG. 3, the radar system 300 includes a synthesizer 301 configured to generate chirps, a TX antenna 302 for transmitting chirps generated by the synthesizer 301, and a RX antenna 303 for receiving chirp reflections in response to transmitted chirps. The radar system 300 also includes a mixer 310 that provides input signals 313 to a low-pass filter 320, where each input signal 313 includes information regarding the chirp 314 and any chirp reflections 316. The graph 312 shows frequency relative to time to illustrate a chirp 314 and a chirp reflection 316 (i.e., the chirp 314 is a signal with increasing frequency as a function of time, and the chirp reflection 316 is a delayed version of the chirp 314). In some examples, the mixer 310 outputs a sinusoid wave with a frequency equal to the difference between the instantaneous frequency of the chirp 314 and the instantaneous frequency of the chirp reflection 316 at time τ in graph 312. In other words, the phase of the sinusoid wave generated by the mixer 310 is equal to the difference of the phase of the chirp 314 and the phase of the chirp reflection 316.

[0025] Each of the input signals 313 is filtered by the low-pass filter 320 and is digitized by an analog-to-digital converter (ADC) 330. The output of the ADC 330 is a digitized sense signal 333, where each digitized sense signal 333 includes chirp and chirp reflection information. Each digitized sense signal 333 is provided to an FFT engine 340 to perform FFT in the first dimension with a combined bit-reversal and memory transpose arrangement 342. FFT engine 340 takes in data in serial order of indices and outputs data in bit-reversed order of indices. In at least some examples, the combined bit-reversal and memory transpose arrangement 342 includes memory devices, DMA and circular shift components to provide bit-reverse address ordered and memory transposed multi-dimensional FFT output samples for use by a processor 350 to determine object parameters for one or more objects that cause chirp reflections 316. Example object parameters include object position, object velocity, and object angle relative to a radar origin (the location of the radar system 300).

[0026] FIG. 4 is a block diagram showing part of a radar data processing system 400 in accordance with various examples. In the example of FIG. 4, the radar data processing system 400 includes the FFT engine 340 and the processor 350 introduced in FIG. 3. Also represented in FIG. 4 is a graph 430 showing example radar data processing operations related to frame 432 and intra-frame time 434. More specifically, radar data processing operations related to four chirps (C1-C4) of frame 432 are represented, where the radar data processing operations involve obtaining FFT output

samples **410** by the FFT engine **340**. More specifically, during frame **432**, the FFT engine **340** organizes the FFT output samples **410** into range bins **411** indexed by chirp number. These FFT output samples **410** are stored in a radar data memory (referred to a second memory device herein) in a bit-reversed and memory transposed arrangement. The FFT output samples **410** are used to perform a range analysis **411**. Example operations for the range analysis **411** involve using at least some of the FFT output samples **410** for individual chirps to determine the range of one or more objects relative to a radar origin.

[0027] Thereafter, during the intra-frame time **434**, the FFT output samples **410** are used to perform a Doppler analysis **421** by using the FFT engine to perform FFT operation across all Doppler/Velocity bins to perform FFT in the second dimension. A third FFT operation can also be performed across FFT output samples from multiple antennas in the third dimension to determine the object angle relative to the radar origin. As previously noted in the discussion for FIG. 3, the FFT engine **340** includes a combined bit-reversal and memory transpose arrangement **342** (not shown in FIG. 4) to organize the FFT output samples **410** in the radar data memory, where combined bit-reversal and memory transpose operations of the FFT engine **340** reduce radar data processing latency compared to the FFT engines introduced in FIGS. 1 and 2 for multi-dimensional FFTs

[0028] In accordance with at least some examples, the FFT engine **340** provides the FFT output samples **410** used for range analysis **411**, Doppler analysis **421** and angle analysis (not shown in figure) by using the FFT engine three times to perform the FFT operations across the three dimensions. More specifically, the FFT engine **340** performs combined bit-reversal and memory transpose operations to store the FFT output samples **410** in a memory accessible to the processor **350** and DMA logic **440**. In some examples, the DMA logic is configured to transfer data from a first memory device to a second memory after the combined bit-reversal and transpose operations. Also, in some examples, the DMA logic is configured to transfer data from a second memory device to the FFT engine **340** to perform multi-dimensional FFT operations.

[0029] In some examples, the processor **350** performs range analysis **411**, Doppler analysis **421**, and angle analysis operations using the final FFT output samples **410**. The process of acquiring chirp and chirp reflection information and of performing radar data processing operations continues for each subsequent frame **436** and subsequent intra-frame time (not shown), where each set of chirps and related reflections provides updated information from which to determine updated object parameters such as object position, object velocity, and object angle relative to a radar origin.

[0030] FIG. 5 is a block diagram showing an FFT engine **500** in accordance with various examples. In some examples, the FFT engine **500** corresponds to the FFT engine **340** of FIG. 3 (used for radar data processing). In other examples, the FFT engine **500** is used for other sense signal analysis (e.g., image analysis), including multi-dimensional FFT operations. As shown in FIG. 5, the FFT engine **500** includes FFT logic **501**, which provides K FFT output samples at a time, and N total FFT output samples for each digitized sense signal **333**. In the example of FIGS. 5, K=4 and N=16. In other examples, the FFT logic **501** varies with regard to the values of K and N, where K and N are integers, and where N is a multiple of K.

[0031] In some examples, the FFT engine topology **500** uses the FFT logic **501** to perform multi-dimensional FFT operations on a multi-dimensional array of size (n.sub.1, n.sub.2, . . . n.sub.m) having m dimensions. In one example involving a 2D array, the FFT logic **501** executes a 1D FFT on n.sub.1 rows of the input array, followed by a 1D FFT on n.sub.2 columns of the first FFT output array. Also, in some real-time applications, where high throughput is desired, the FFT logic **501** is configured to process several samples of an input sequence in parallel and to provide multiple outputs in parallel. For example, if the FFT logic **501** is a 4X-FFT, 4 outputs are generated every cycle. To achieve a high-throughput FFT (e.g., 4X-FFT) for the FFT logic **501**, different parallel pipelined FFT architectures are available and can be implemented in hardware.

[0032] Returning to FIG. 5, the FFT engine **500** also comprises a first memory device **520** with a

plurality of banks (B0-B3). In some examples, the number of banks for the first memory device **520** equals K (the number of FFT output samples provided by the FFT logic **501** at a time), which is 4 in the example of FIG. 5. The FFT engine **500** also comprises a bit-reversed address generator **511**, direct memory access (DMA) logic **550**, and a second memory device **540**. The FFT engine **500** also comprises a first set of circular shift components **502** and a second set of circular shift components **530**.

[0033] In operation, the first set of circular shift components **502** are configured to perform shifts between banks of the first memory device **520** when writing FFT output samples in bit-reversed address order generated by the FFT logic **501** to the first memory device **520**. More specifically, the first set of circular shift components **502** are configured to write FFT output samples corresponding to each digitized sense signal **333** across the plurality of banks of the first memory device **520**. In some examples, the first set of circular shift components **502** include a set of write data multiplexers **504A** coupled between the FFT logic **501** and the first memory device **520**. The first set of circular shift components **502** also include a set of address multiplexers **504B** coupled between the bit-reversed address generator **511** and the first memory device **520**, where the set of write data multiplexers **504A** and the set of address multiplexers **504B** performs writes to B0-B3 with circular right shift of 1 bank based on a FFT index number (FFT_INDEX) or identifier that increments after the completion of an FFT operation in a particular dimension. In some examples, the FFT index number is a modulo X number, where X is the number of banks in the first memory. The FFT index number is reset to '0' at the start of a new dimension

[0034] More specifically, the set of address multiplexers **504B** are configured to use addresses from the bit-reversed address generator **511** to write FFT output samples for a given FFT operation to B0-B3 of the first memory device **520** before circularly shifting writes to store FFT output samples for another FFT operation in B0-B3 of the first memory device **520**. In some examples, the set of write data multiplexers **504A** and the set of address multiplexers **504B** operate to write K FFT output samples at a time to B0-B3 of the first memory device **520**. Also, in some examples, an FFT index number (FFT_index) is provided by a modulo K counter and is used to control operations of the set of write data multiplexers **504A** and operations of the set of address multiplexers **504B**. More specifically, in one example, the bit-reversed address generation module **511** generates an address between 0 and $(N/K-1)$. In the example of FIG. 5, the bit-reversed address generation module **511** generates a number between 0 and 3. Based on the number generated by the bit-reversed address generation module **511** and the operations of address adjustment logic **506**, the set of address multiplexers **504B** generate K addresses, where the addresses are bit-reversed relative to the FFT output samples provided by the FFT logic **501** and are spaced apart by $N/4$ in the example of FIG. 5 (the address adjustment logic **506** provides spaced addresses as needed).

[0035] For the example of FIG. 5, the FFT logic **501** is represented as providing an array of FFT output samples $X[k]$, $X[k+N/4]$, $X[k+2N/4]$, $X[k+3N/4]$. In some examples, k varies from 0-3 and $N=16$. In such case, the set of write data multiplexers **504A** shifts the FFT output samples $X[k]$, $X[k+N/4]$, $X[k+2N/4]$, $X[k+3N/4]$, so that one of four available FFT output samples is stored in a different one of B0-B3. For instance, each FFT output sample $X[k]$ is stored in bank B0, each FFT output sample $X[k+N/4]$ is stored in bank B1, each FFT output sample $X[k+2N/4]$ is stored in bank B2, and each FFT output sample $X[k+3N/4]$ is stored in bank B3, where the value of k is 0, 1, 2, and 3 as a function of time. In the following FFT operations, the four FFT output data words are circularly shifted and each FFT output sample $X[k]$ is stored in bank B1, each FFT output sample $X[k+N/4]$ is stored in bank B2, each FFT output sample $X[k+2N/4]$ is stored in bank B3, and each FFT output sample $X[k+3N/4]$ is stored in bank B0, and so on

[0036] Meanwhile, the second set of circular shift components **530** are configured to perform shift between banks of the first memory device **520** when reading FFT output samples in linear address order from the first memory device **520** for storage in the second memory device **540**. More specifically, the second set of circular shift components **530** are configured to read FFT output

samples from the first memory device **520** for storage in the second memory device **540** such that FFT output samples corresponding to each of the digitized sense signals **333** are read in sequential order from the plurality of banks of the first memory device **520**. In some examples, the second set of circular shift components **530** include a set of read data multiplexers coupled between the first memory device **520** and the DMA logic **550** coupled to the second memory device **540**. As an example, the read data multiplexers are configured to circularly shift reads of FFT output samples from one of B0-B3 of the first memory device **520** to another of B0-B3 at a predetermined rate equal to N/K read cycles to read FFT output samples corresponding to each digitized sense signal **333** in transposed order. For the FFT engine **500** of FIG. 5, at least some of the bit-reversal operations provided by the first set of circular shift components **502** and at least some of the memory transpose operations provided by the second set of circular shift components **530** are combined.

[0037] With the FFT engine **500**, the second memory device **540** stores transposed FFT output samples in linear address order for multiple FFT outputs in any given dimension, where these FFT output samples are available for further FFT processing in another dimension or further analysis by the processor (see FIG. 4). In a radar data processing examples, the FFT output samples stored by the second memory device **540** are accessed by a processor (e.g., the processor **350**) to determine object parameters such as object position, object velocity, and object angle relative to a radar origin.

[0038] With the FFT engine **500**, the first set of circular shift components **502** are configured to write FFT output samples corresponding to each digital input signal across the plurality of banks (B0-B3) of the local memory **520** in bit-reversed address order. Also, the second set of circular shift components **530** are configured to read FFT output samples from the local memory **520** for storage in the data memory **540** such that FFT output samples corresponding to each of the digital input signals are read in linear order from the plurality of banks of the local memory **520**. In some examples, the first set of circular shift components **502** include a set of write data multiplexers **504A** coupled between the FFT logic **501** and the local memory **520**. In some examples, the set of write data multiplexers **504A** includes K write data multiplexers **504A** configured to write each one of the K FFT output samples to one of the plurality of banks (B0-B3) of the local memory **520** before shifting to another of the plurality of banks of the first local memory **520** for the next FFT operation, where K is at least 2. More generically, the number of banks for the local memory **520** is equal to X , where X is at least equal to K . Also, in some examples, an index number is used to control shift operations of the first set of circular shift components **502**, where the index number is incremented after each FFT operation and is a modulo X number. Also, in some examples, the first set of circular shift components **502** include a set of address multiplexers **504B** coupled between the bit-reversed address generator **511** and the first memory device **520**. In some examples, the set of address multiplexers **504B** are configured to use K addresses from the bit-reversed address generator **511** to write to one of the plurality of banks (B0-B3) of the local memory **520** before shifting to another of the plurality of banks of the local memory **520** for the next FFT operation, where K is at least 2.

[0039] In some examples, the bit-reversed address generator **511** is configured to generate addresses from 0 to $(N/4)-1$ in bit-reversed order for one bank of the local memory **520**. Also, the bit-reversed address generator **511** is configured to generate addresses for other banks of the local memory **520** by adding an address offset of N/K to a previous bank address, where N is the size of the FFT, and where K is the number of parallel inputs and outputs of the FFT logic **501**.

[0040] In some examples, the second set of circular shift components **530** include a set of read data multiplexers coupled between the local memory **520** and DMA logic **550** coupled to the data memory **540**. In some examples, read data multiplexers of the second set of circular shift components **530** are configured to shift reads of FFT output samples from one of the plurality of banks of the local memory **520** to another of the plurality of banks of the local memory **520** at a

predetermined rate to read FFT output samples for each digital input signal in order. In some examples, read data multiplexers of the second set of circular shift components **530** are reconfigured at a predetermined rate equal to N/K read cycles, where N is the size of the FFT logic, and K is the number of parallel inputs and parallel outputs of the FFT logic **501**.

[0041] In some examples, the local memory **520** has a capacity to store at least K FFT output array size (i.e., the total capacity of the local memory **520** will be equal to $K \times N$ times FFT output sample size) and has at least K banks. Also, the data memory **540** has a capacity that is greater than or equal to local memory **520**, and where K is the number of parallel inputs and parallel outputs of the FFT logic **501**. Also, in some examples, the DMA logic **550** is configured to transfer data from local memory **520** to the data memory **540** after the combined bit-reversal and transpose operations, and wherein the DMA logic **550** is also configured to transfer data from the data memory **540** to the FFT logic **501** to perform multi-dimensional FFT operations.

[0042] FIGS. **6A-6D** are a block diagrams illustrating example operations of the FFT engine **500** of FIG. **5** in accordance with various examples. In FIG. **6A**, the represented operations **600A** include writing some of a set of FFT output samples **601** for a first FFT to B0-B3 of the first memory device **520**. More specifically, when FFT output samples with index values 0, 4, 8, 12 ($k=0$) are provided by the FFT logic **501** first, the first set of circular shift components **502** is used to write these FFT output samples to B0-B3 based on an identifier and/or FFT index number (FFT_index). For the represented operations **600A** involving FFT output samples for a first FFT, the FFT output sample with index value 0 is written to the first row address of B0, the FFT output sample with index value 4 is written to the fifth row address of B1, the FFT output sample with index value 8 is written to the ninth row address of B2, and the FFT output sample with index value 12 is written to the thirteenth row address of B3.

[0043] In FIG. **6B**, the represented operations **600B** include writing other FFT output samples from the second set of FFT output samples **601** for the first FFT to B0-B3 of the first memory device **520**. More specifically, the FFT output samples with index values 2, 6, 10, 14 ($k=2$) are provided by the FFT logic **501** next, and the first set of circular shift components **502** is used to write these FFT output samples to B0-B3 based on an identifier and/or FFT index number (FFT_index). For the represented operations **600B** involving FFT output samples for a first FFT, the FFT output sample with index value 2 is written to third row address of B0, the FFT output sample with index value 6 is written to the seventh row address of B1, the FFT output sample with index value 10 is written to the eleventh row address of B2, and the FFT output sample with index value 14 is written to the fifteenth row address of B3.

[0044] While not shown in FIG. **6B**, the process of writing other FFT output samples from the set of FFT output samples **601** for the first FFT to B0-B3 of the first memory device **520** continues with $k=1$. More specifically, first set of circular shift components **502** is used to write the FFT output sample with the index value 1 to the second row address of B0, to write the FFT output sample with the index value 5 to the sixth row address of B1, to write the FFT output sample with the index value 9 to the tenth row address of B2, and to write the FFT output sample with the index value **13** to the fourteenth row address of B3. Thereafter, the process of writing other FFT output samples from the set of FFT output samples **601** for the first FFT to B0-B3 of the first memory device **520** continues with $k=3$. More specifically, the first set of circular shift components **502** is used to write the FFT output sample with the index value 3 to the fourth row address of B0, to write the FFT output sample with the index value 7 to the eighth row address of B1, to write the FFT output sample with the index value 11 to the twelfth row address of B2, to write the FFT output sample with the index value 15 to the sixteenth row address of B3.

[0045] In FIG. **6C**, the represented operations **600C** include writing FFT output samples from a set of FFT output samples **602** for a second FFT to B0-B3 of the first memory device **520**. More specifically, the FFT output samples with index values 0, 4, 8, 12 ($k=0$) are provided by the FFT logic **501** first, and the first set of circular shift components **502** is used to write these FFT output

samples to B0-B3 based on an identifier and/or FFT index number (FFT_index). For the represented operations **600C**, the first set of circular shift components **502** FFT is used to write the FFT output sample with index value 12 to the thirteenth row address of B0, to write the FFT output sample with index value 0 to the first row address of B1, to write the FFT output sample with index value 4 to the fifth row address of B2, and to write the FFT output sample with index value 8 to the ninth row address of B3.

[0046] While not shown in FIG. **6C**, the process of writing other FFT output samples from second set of FFT output samples **602** for the second FFT to B0-B3 of the first memory device **520** continues with k=2. When k=2, the FFT output samples with index values 2, 6, 10, 14 are provided by the FFT logic **501**, and the first set of circular shift components **502** is used to write these FFT output samples to B0-B3 based on an identifier and/or FFT index number (FFT_index). More specifically, the first set of circular shift components **502** is used to write the FFT output sample with index value 14 to the fifteenth row address of B0, to write the FFT output sample with index value 2 to the third row address of B1, to write the FFT output sample with index value 6 to the seventh row address of B2, and to write the FFT output sample with index value 10 is written to the eleventh row address of B3.

[0047] While not shown in FIG. **6C**, the process of writing other FFT output samples from the third set of FFT output samples **602** for the second FFT to B0-B3 of the first memory device **520** continues with k=1. When k=1, the FFT output samples with index values 1, 5, 9, and 13 are provided by the FFT logic **501** and the first set of circular shift components **502** is used to write these FFT output samples to B0-B3 based on an identifier and/or FFT index number (FFT_index). More specifically, the first set of circular shift components **502** is used to write the FFT output sample with the index value 13 to the fourteenth row address of B0, to write the FFT output sample with the index value 1 to the second row address of B1, to write the FFT output sample with the index value 5 to the sixth row address of B2, and to write the FFT output sample with the index value 9 to the tenth row address of B3.

[0048] While not shown in FIG. **6C**, the process of writing other FFT output samples from the fourth set of FFT output samples **602** for the second FFT to B0-B3 of the first memory device **520** continues with k=3. When k=3, the FFT output samples with index values 3, 7, 11, and 15 are provided by the FFT logic **501** and the first set of circular shift components **502** is used to write these FFT output samples to B0-B3 based on an identifier and/or FFT index number (FFT_index). More specifically, the first set of circular shift components **502** is used to write the FFT output sample with the index value 15 to the sixteenth row address of B0, to write the FFT output sample with the index value 3 to the fourth row address of B1, to write the FFT output sample with the index value 7 to the eighth row address of B2, and to write the FFT output sample with the index value 11 to the twelfth row address of B3.

[0049] While not shown in FIG. **6C**, the process of writing other FFT output samples from other sets of FFT output samples **602** (e.g., for a third FFT and a fourth FFT) continues until B0-B3 are filled with the FFT output samples for four FFTs, where the FFT output samples are distributed across B0-B3.

[0050] FIG. **6D** shows represented operations **600D** to read FFT output samples stored in B0-B3 for storage in the second memory device **540** using the second set of circular shift components **530** and DMA logic **550**. As shown in FIG. **6D**, the FFT output samples for four FFTs are distributed across B0-B3 such that same address location in all 4 banks contain the same FFT output index. Accordingly, read operations performed in linear order using the second set of circular shift components **530** shift left by 1 bank when reading the FFT output samples after every N/K read cycles More specifically, read operations **604** for FFT output samples involve reading the same memory addresses across B0, B1, B2, B3 in linear order. With the second set of circular left shift components **530**, read operations to store FFT output samples from all 4 FFTs in the first memory into the second memory device **540** result in FFT output samples stored in the second memory

device **540** having a memory transposed format **620**.

[0051] Writing FFT outputs into the first memory device using a bit-reversed address and reading out in linear order before storing into the second device is performed to arrange the FFT outputs in linear order inside the second memory device. Along with this bit-reversal operation, the combination of first and second set of circular shift components also allows writing the FFT outputs across multiple FFTs in transpose order into the second memory device without incurring any additional latency for transpose operation

[0052] In some examples, with the FFT engine **500**, operations (e.g., the operations **600A**, **600B**, **600C**, **600D** represented in FIGS. **6A-6D**) are repeated for consecutive K blocks of FFTs in one dimension, after which the FFT output samples are stored in a linear and transposed order in the second memory device **540**. These transposed FFT outputs can then be accessed in linear order to perform the second set of FFT computations in the second dimension and so on. In a radar data processing examples, the final FFT output samples after multi-dimensional FFT computations are accessed by a processor (e.g., the processor **350**) to determine object parameters such as object position, object velocity, and object angle relative to a radar origin.

[0053] FIG. **7** is a block diagram of a radar data processor device **700** in accordance with various examples. In different examples, the radar data processor device **700** corresponds to an integrated circuit, a multi-die module, a printed circuit board (PCB) with components, and/or other radar data processing device options. As shown, the radar data processor device **700** includes a synthesizer **301A** (an example of the synthesizer **301** in FIG. **3**), a transmitter antenna **302A** (an example of the transmitter antenna **302** in FIG. **3**), a plurality of receiver antennas **303A-303N** (examples of the receiver antenna **303** in FIG. **3**), and mixers **310A-310N** (examples of the mixer **310** in FIG. **3**). The output of the mixers **310A-310N** are used as input signals **313A-313N** (examples of the input signal **313**) that include chirp and chirp reflection information. In some examples, the transmitter antenna **302A** and/or the receiver antennas **303A-303N** are separate from the other components represented for the radar data processor device **700** (e.g., the transmitter antenna **302A** and/or the receiver antennas **303A-303N** are on-chip or off-chip in different examples). Also, it should be appreciated that the receiver antennas **303A-303N** are spaced from each other. Also, in some examples, the transmitter antenna **302A** is usable as one of the receiver antennas **303A-303N** and vice versa (with appropriate circuitry to connect/disconnect an antenna to other components as desired). At least one antenna is needed, with multiple antennas providing redundancy, angle information, and/or other radar system options.

[0054] The radar data processor device **700** also includes an ADC/filter **702A-702N** for each of the input signals **313A-313N** to filter and digitize the input signal **313A-313N**. As shown, the radar data processor device **700** also includes an FFT engine **340A** (an example of the FFT engine **340** in FIG. **3**) configured to perform FFT operations as described herein for each of the digitized sense signal **333A-333N**. With FFT output samples for different receiver antennas **303A-303N**, object parameters are determined based on bit-reversed and memory transposed FFT output samples as described herein. More specifically, the FFT engine **340A** includes a combined bit-reversal and memory transpose arrangement **342A** (an example of the combined bit-reversal and memory transpose arrangement **342** in FIG. **3**). In some examples, the combined bit-reversal and memory transpose arrangement **342A** includes a first memory device with multiple banks (e.g., the first memory device **520** of FIG. **5**), address generation logic (e.g., the address generation logic **506** of FIG. **5**), circular shift components before and after the first memory device (e.g., the first and second set of circular shift components **502** and **530** of FIG. **5** to shift writes and reads according to a identifier or FFT index number), and a second memory device (e.g., the second memory device **540** in FIG. **5**) to store FFT output samples after bit-reversal addressing and memory transpose operations. In other examples, an FFT engine, such as the FFT engine **340** (having combined transpose and bit reversal re-ordering), can also be sold as a standalone FFT engine (e.g., in a digital only chip) and/or can be combined with other components.

[0055] FIG. 8 is a flowchart showing an FFT engine method **800** in accordance with various examples. As shown, the method **800** comprises performing parallel pipeline FFT operations to generate FFT output samples at block **802**. At block **804**, writing operations are performed to write the FFT output samples to a first memory device, wherein the writing operations involve shifting between different banks of the first memory device and writing the FFT output samples in bit-reversed address order. At block **806**, reading operations are performed to read the FFT output samples from the first memory device, wherein the reading operations involve shifting between different banks of the first memory device and reading FFT output samples in linear address order for storage of transposed FFT output samples in a second memory device. If another dimension is to be analyzed (determination block **808**), the transposed FFT output samples are provided to block **802** for analysis in another dimension. In some examples, multi-dimensional analysis involves use of DMA logic to convey transposed FFT output samples to FFT logic associated with block **802**. If another dimension is not to be analyzed (determination block **808**), the transposed FFT output samples on the second memory device are used to perform at least one subsequent operation at block **810**.

[0056] In some examples, the method **800** is used to analyze sense signals (e.g., a radar chirp and radar chirp reflections, or image data) in one or more dimensions. In some examples, the method **800** also including receiving a sense signal, wherein the performing parallel pipeline FFT operations generates FFT output samples related to the sense signal. In such case, the transposed FFT output samples provided at block **806** are used to analyze an object associated with the sense signal. In some examples, the sense signal to be analyzed in or more dimensions includes radar chirp and radar chirp reflection information. In some examples, a processor is able to use transposed FFT output samples for one or more dimensions to detect object parameters such as object position relative to a radar system position, object speed relative to the radar system position, and/or object angle relative to a radar system position.

[0057] Certain terms have been used throughout this description and claims to refer to particular system components. As one skilled in the art will appreciate, different parties may refer to a component by different names. This document does not intend to distinguish between components that differ in name but not function. In this disclosure and claims, the terms “including” and “comprising” are used in an open-ended fashion, and thus should be interpreted to mean “including, but not limited to . . .” Also, the term “couple” or “couples” is intended to mean either an indirect or direct wired or wireless connection. Thus, if a first device couples to a second device, that connection may be through a direct connection or through an indirect connection via other devices and connections. The recitation “based on” is intended to mean “based at least in part on.” Therefore, if X is based on Y, X may be a function of Y and any number of other factors.

[0058] The above discussion is meant to be illustrative of the principles and various embodiments of the present invention. Numerous variations and modifications will become apparent to those skilled in the art once the above disclosure is fully appreciated. It is intended that the following claims be interpreted to embrace all such variations and modifications.

Claims

1. A data processing device comprising: logic circuitry configurable to receive a plurality of digital signals and perform transform operations to generate N output samples for each of the plurality of digital signals, k output samples at a time, wherein k is an integer of 2 or greater and N is an integer multiple of k; a first memory device with k banks; a second memory device; and circular shift circuitry that includes data multiplexers and address multiplexers configurable to write respective sets of k output samples of the N output samples to the k banks of the first memory device wherein, for each set of k output samples, the circular shift circuitry is further configurable to: generate, using the address multiplexers, k addresses that are bit-reversed with respect to the k output

samples, and write a respective one of the k output samples at an indexed ordered position in a respective one of the k banks; wherein the data processing device is configurable to read the output samples in linear address order from the k banks of the first memory device and store the output samples in a transposed format in the second memory device.

2. The data processing device of claim 1, wherein the second memory device includes k banks.
3. The data processing device of claim 2, wherein, to store the output samples in a transposed format in the second memory device, the data processing device is further configurable to store each group of N output samples in a respective one of the k banks of the second memory device.
4. The data processing device of claim 1, further comprising a bit-reversed address generator coupled to the first memory device, the bit-reversed address generator configurable to provide a bit-reversed address order to the address multiplexers for writing the output samples in different memory banks of the k banks of the first memory device.
5. The data processing device of claim 1, further comprising a set of circular shift components coupled to the first memory device and to the second memory device, the set of circular shift components configurable to shift between the k banks of the first memory device to read the output samples in linear address order from the k banks of the first memory device and store the output samples in a transposed format in the second memory device.
6. The data processing device of claim 1, wherein the data processing device is part of a radar system that includes receiving circuitry configurable to receive radar signals and to generate the plurality of digital signals based on the radar signals.
7. The data processing device of claim 6, wherein each of the of the plurality of digital signals includes radar chirp transmission information and radar chirp reflection information.
8. The data processing device of claim 6, further comprising processing circuitry configurable to use the output samples stored in transposed format in the second memory device to determine at least one of a distance of an object from the radar system, a velocity of an object relative to the radar system, and an angle of an object relative to the radar system.
9. The data processing device of claim 1, wherein the circular shift circuitry is further configurable to write respective sets of k output samples of the N output samples to the k banks of the first memory device by shifting between the k banks of the first memory device and writing the N output samples to the k banks of the first memory device in bit-reversed order.
10. The data processing device of claim 1, wherein to read the output samples in linear address order from the k banks of the first memory device, the data processing device is further configurable to shift between the k banks of the first memory device.
11. The data processing device of claim 1, further comprising a circular shifter configurable to read the output samples in linear address order from the k banks the first memory device.
12. The data processing device of claim 1, wherein $N=16$ and $k=4$.
13. The data processing device of claim 1, wherein, to read the output samples from the k banks of the first memory device, the data processing device is further configurable to shift from one of the k banks to another of the k banks at a rate of N/k .
14. A system comprising: logic circuitry configurable to receive a plurality of digital signals and perform transform operations to generate N output samples for each of the plurality of digital signals, k output samples at a time, wherein k is an integer of 2 or greater and N is an integer multiple of k; a first memory device with k banks; a second memory device; circular shift circuitry that includes data multiplexers and address multiplexers configurable to perform write operations to write respective sets of k output samples of the N output samples to the k banks of a first memory device, wherein, for each set of k output samples, the circular shift circuitry is configurable to write a respective one of the k output samples in a respective one of the k banks, in which each of a second to a last of the k output samples is offset in position with respect to a position of a preceding one of the k output samples by k; and a circular shifter configurable to read the output samples in linear address order from the k banks of the first memory device and store the

output samples in a transposed format in the second memory device.

15. The system of claim 14, wherein the second memory device includes k banks, and after the circular shifter is further configurable to store respective groups of N output samples in respective banks of the k banks of the second memory device.

16. The system of claim 14, further comprising a bit-reversed address generator coupled to the first memory device, the bit-reversed address generator configurable to provide a bit-reversed address order to the address multiplexers for writing the output samples in different memory banks of the k banks of the first memory device.

17. The system of claim 14, further comprising receiving circuitry configurable to receive radar signals and to generate the plurality of digital signals based on the radar signals.

18. The system of claim 17, wherein each of the of the plurality of digital signals includes radar chirp transmission information and radar chirp reflection information.

19. The system of claim 17, further comprising processing circuitry configurable to use the output samples stored in transposed format in the second memory device to determine at least one of a distance of an object from the system, a velocity of an object relative to the system, and an angle of an object relative to the system.
